Parametric test

t-test(Sample, Pooled or Unpaired and Paired) ,
ANOVA (One way and Two way),
Least Significance difference

Dr. Pramila Chaubey
cpramil@su.edu.sa

Parametric test

parametric test is a test whose model *specifies certain conditions* about the parameters of the population from which the sample are drown.

- used to evaluate hypotheses

- Types of Parametric test
  - t-test
  - Z-test
  - F-test
  - Analysis of variance (ANOVA)

- Parametric tests assume a normal distribution of values

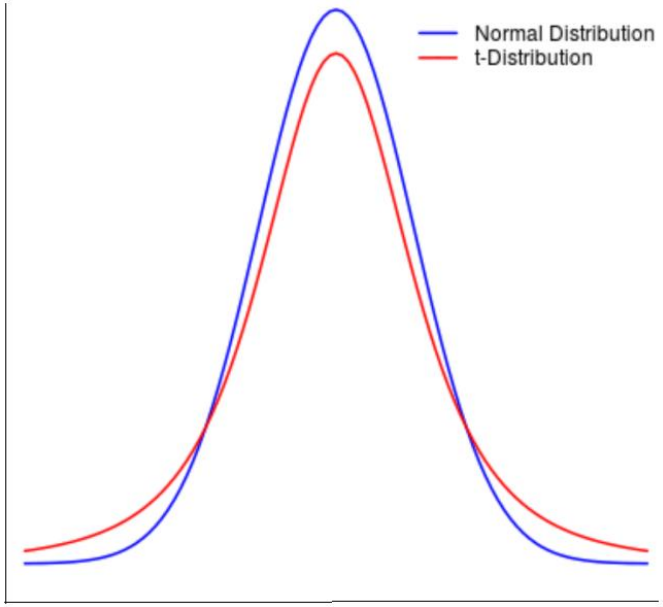
T test

Synonym: Student's t-distribution

Defination :

t distribution is the probability distribution that is used to estimate population parameters when the sample size is small and the population variance is unknown.

- is a type of probability distribution that is similar to the normal distribution with its bell shape but has heavier tails.
- T-tests are used in statistics to estimate significance (hypothesis testing)
- The t-test determines either whether the group mean and the mean of the population or two group means differ statistically.



T test

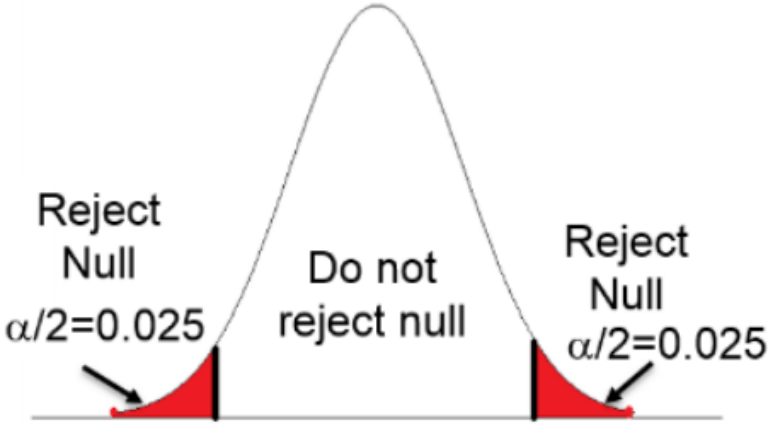
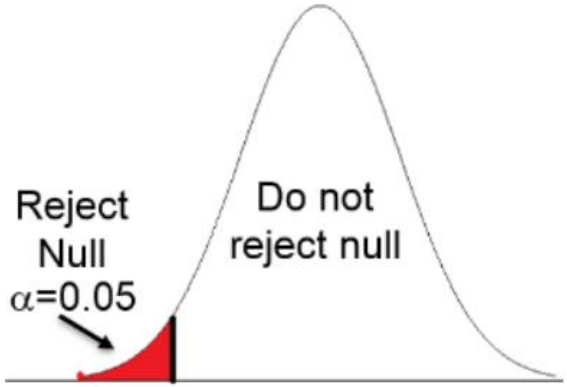
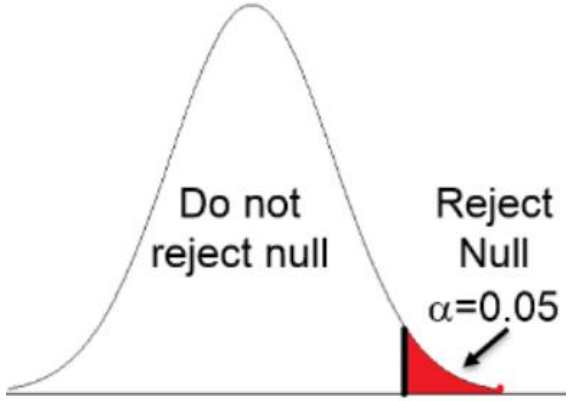
3 types of t-test

One-sample test	Unpaired sample test	paired sample test
There is a difference btwn a group and the population (hypothesized value of population mean)	There is a difference btwn two groups	There is a difference in a group btwn *two points* at different occasion
d.f. = N-1	d.f. = $N_1 + N_2 - 2$	d.f. = N-1
	Also called as *Pooled t test* *independent* t test	Also called as *Difference t-Test* *dependent* t test
Used in studies of Label potency : Average tablet potency of N tablets	Used for a parallel-groups design : Active drug compared to a standard drug or placebo, Tablet dissolution, potency, weight, etc., from two batches	Used in Stability studies : Assays over time from material from same container crossover study : Each patient takes each drug on different occasions
$t = \frac{\lvert \bar{X} - \mu_0 \rvert}{S/\sqrt{N}}$	$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$	$t = \frac{\bar{\delta} - \Delta}{S/\sqrt{N}}$

T test

Two-tailed	One-tailed
Also called as non-directional, two sided tail	Also called as directional, one sided tail
Used when $H_1 : \mu_1 \neq \mu_2$	Used when $H_1 : \mu_1 > \mu_2$ right-tailed $H_1 : \mu_1 < \mu_2$ left-tailed
Alternative hypothesis H_1: The mean value of the population is not equal to the specified value	**Alternative hypothesis H_1**: The mean value of the population is smaller (or larger) than the specified values.
level of significance equally distributed with half value on both tails	level of significance distributed on either side tail

T test

Two-tailed	One-tailed
For α=0.05 	For α=0.05  left-tailed or  right-tailed

One-sample test

Case I

- After A large amount of data was collected for the content of drug in the tablet formulation during a period of several years , The manufacturing process showed an average potency of 5.01 mg and a standard deviation of 0.11, *both values considered to be equal to the true process parameters*.
- A new batch was made with a modification of the usual manufacturing procedure. And assay of 20 tablets give bellow..

Results of 20 Single-Tablet Assays from a Modification of a Process with a Historical Mean of 5.01 mg σ (historical) = 0.11

5.13	5.04	5.09	5.00
4.98	5.03	5.01	4.99
5.20	5.08	4.96	5.18
5.08	5.06	5.02	5.24
4.99	5.17	5.06	5.00
	$\overline{X} = 5.0655$ mg	$S = 0.0806$	

One-sample test

- The objective is…

  …… to determine if the process modification results in a change of average potency from the process average of 5.01, the value of under the null hypothesis.

The steps for designing and analyzing this experiment are as follows:

1. *Careful planning*

- An appropriate experimental design
- the routine quality control content uniformity assay was the determined with 20 tablets were chosen at random from the newly manufactured batch.

One-sample test

2. The *null hypothesis* and *alternative hypothesis*

two sided *hypothesis testing*

$$H_0 : \mu = \mu_0$$
$$H_a : \mu \neq \mu_0.$$

- Alternative hypothesis suggest that the average drug potency of the new batch can conceivably be smaller as well as greater than the historical process average of 5.01 mg.

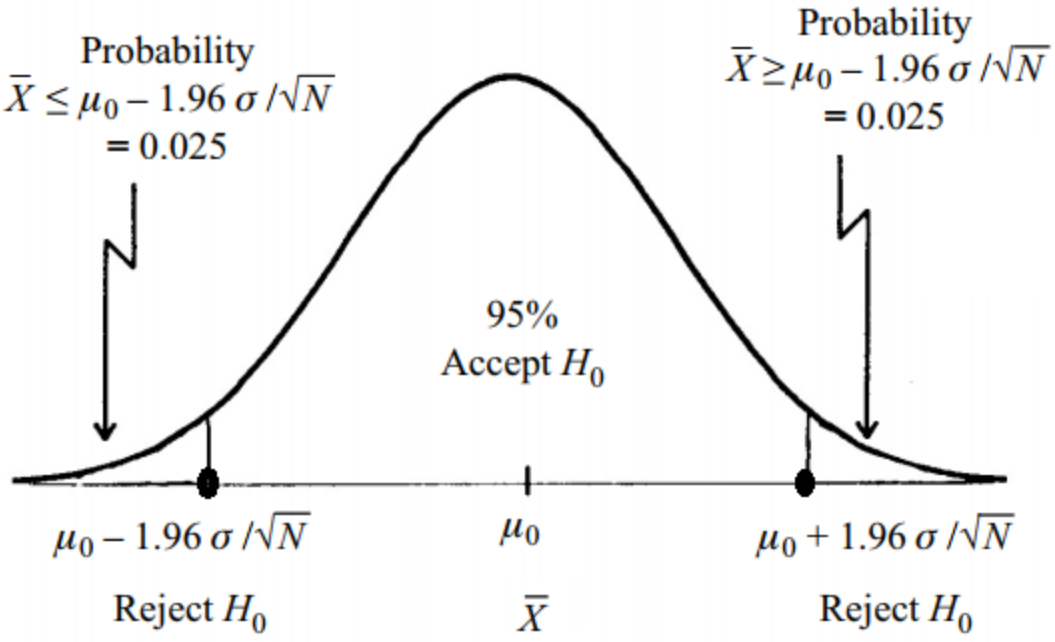
one sided *hypothesis testing*

$$H_0 : \mu \leq 5.01\,\text{mg}$$
$$H_a : \mu > 5.01\,\text{mg}.$$

One-sample test

3. The *level of significance* is specified

- the level of significance represents of type I error , the chance of making a mistake when deciding to reject the null hypothesis.
- level of significance is specified depending on the possible risk ,
- majority cases it is 5% level
  expressed as $p < 0.05$



Z ≤ -1.96 or Z ≥ 1.96 then reject H_0

One-sample test

4. Selection of sample
5. Experiment trials

6. After the experiment is completed , Statistical calculations are
   - mean values are to be compared

One-sample test, population variance known. two-sided test

$$Z = \frac{|\overline{X} - \mu_0|}{\sigma/\sqrt{N}} = \frac{|5.0655 - 5.01|}{0.11/\sqrt{20}} = 2.26$$

here, at 5% level values of $|Z| \geq 1.96$ thus, reject null hypothesis.

Therefore, the new batch can be said to have a potency different from previous batches (in this case, the mean is greater).

One-sample test

One-sample test, variance unknown. two-sided test

- The variance is computed from the experimental data.
- the t distribution is used for calculation with equation….

$$t = \frac{|\overline{X} - \mu_0|}{\sqrt{S^2/N}}$$

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}} = \frac{|5.0655 - 5.01|}{0.0806/\sqrt{20}} = 3.08$$

- The value of t needed for significance for a two-sided test at the 5% level and with d.f. 19 is 2.09 [obtained from **t distribution table**]
- The calculated value is more than table t value i.e. reject null hypothesis

One-sample test

Standard error of mean (SEM)

Synonym: Standard Deviation of the Mean

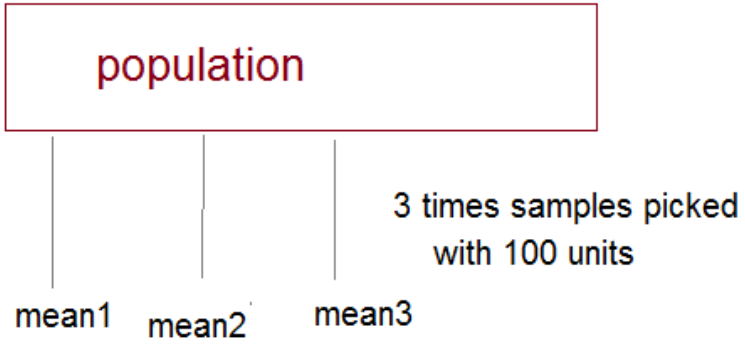
- is a measure of the variability of the sample means
- Standard error gives the accuracy of a sample mean by measuring the sample-to-sample variability of the sample means.
- Is a measure of chance variation , is not an error or mistake

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$

the difference between the means

Standard error of mean

$$S_{\overline{X}} = \frac{S}{\sqrt{N}}.$$



One-sample test

SD	SEM
The s.d. is a measure of the spread of a group of individual observations	SEM is a measure of the variability of the sample means
measures the dispersion from specific data points to the mean	measures the precision of the sample mean
$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{N-1}}$	$S_{\bar{X}} = \frac{S}{\sqrt{N}}$.
Larger than SEM	smaller than the s.d.
Increasing the sample size does not make the SD necessarily larger or smaller; it just becomes a more accurate estimate of the population SD	As the sample size increases, the s.d. of the mean becomes smaller and smaller.

One-sample test

Case II

- A pharmaceutical company places A new compound on an "antihypertensive" screening. Experience has shown that a blood pressure reduction of more than 15 mm Hg in hypertensive animals. since such testing is expensive, the researchers wish to be reasonably sure that statistically the compound truly reduces the blood pressure by more than 15 mm Hg before testing is continued. The level of significance of 10% was chosen

Blood Pressure Reduction Caused by a New Antihypertensive Compound in 10 Animals (mm Hg)

15	12
18	17
14	21
8	16
20	18

One-sample test

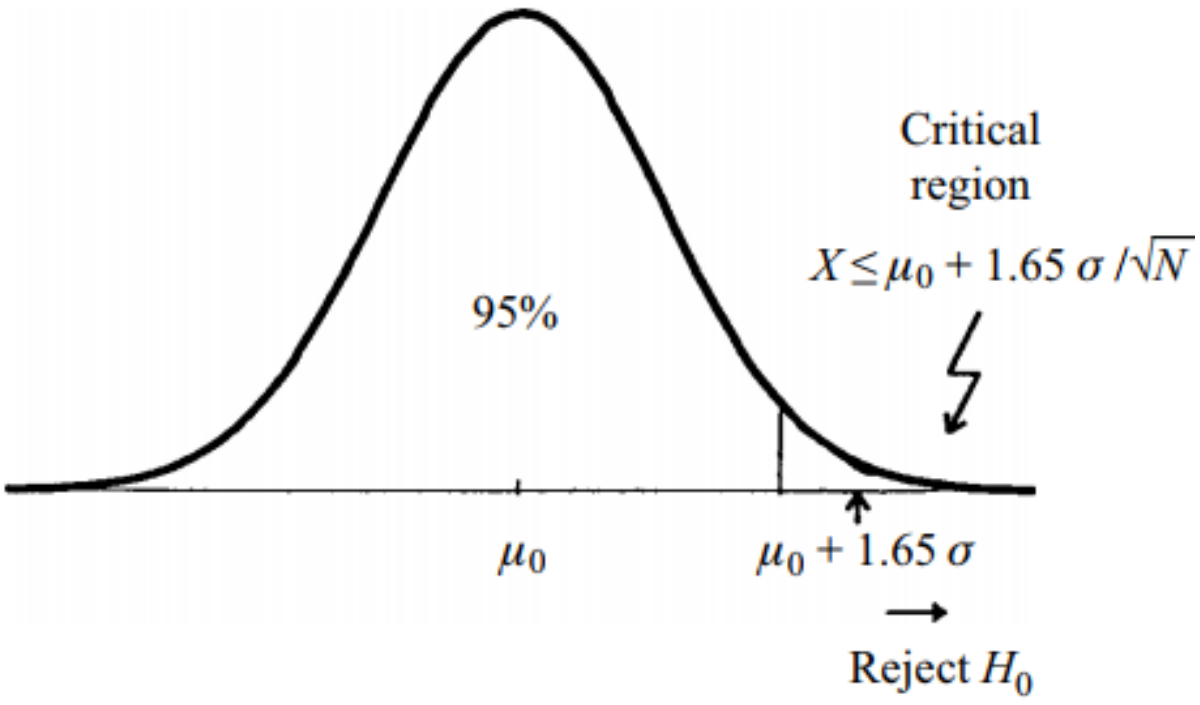
- Hypothesis

$H_o : \mu \leq 15\,\text{mm Hg reduction}$

$H_a : \mu > 15\,\text{mm Hg reduction}$

- This is a *one-sided* test.

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$



One-sample test

- Hypothesis

$H_o : \mu \leq 15\,\text{mm Hg reduction}$

$H_a : \mu > 15\,\text{mm Hg reduction}$

$\overline{X} = 15.9$

$$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{N-1}} = 3.87$$

- This is a *one-sided* test.

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}} = \frac{15.9 - 15}{3.87/\sqrt{10}} = \frac{0.9}{1.22} = 0.74.$$

- The value of t needed for significance for a one-sided test at the 10% level and with d.f. 9 is 1.38 [obtained from **t distribution table**]

- The calculated value is less than table t value i.e. accept null hypothesis
- Therefore, the compound is not sufficiently effective to be considered further.

One-sample test

What difference (reduction) would have been needed to show a significant reduction in this clinical trial ?

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$

$$\overline{X} = \frac{t\ S}{\sqrt{N}} + \mu_0$$

$$= \frac{(1.38)(3.87)}{\sqrt{10}} + 15 = 16.7.$$

- A blood pressure reduction of 16.7 mm Hg or more (the critical region) would have resulted in a significant difference.

Unpaired sample test

Synonym: *Pooled t* test

- Unpaired sample test used for a parallel-groups design

	Two- Independent- Groups Test
Clinical studies	Active drug compared to a standard drug or placebo; treatments given to different persons, one treatment per person
Preclinical studies	Comparison of drugs for efficacy and/or toxicity with treatments given to different animals
Comparison of product attributes from two batches	Tablet dissolution, potency, weight, etc., from two batches

- For *t* test calculation we have two variance for two data set.
- We need to an estimate of the common variance

step 1. first compute the variance of each group

step2. calculate a weighted average of the variances (*pooled variance*)

$$\text{pooled variance} = S_p^2 = \frac{(N_1-1)S_1^2 + (N_2-1)S_2^2}{N_1+N_2-2}$$

Unpaired sample test

EXCERSIZE : Two different formulations of a tablet of a new drug are to be compared with regard to rate of dissolution. Find if the dissolution rates of the two formulations differ. level of significance is 5% level

Percent Dissolution After 15 Minutes for Two Tablet Formulations

Formulation *A*	**Formulation *B***
68	74
84	71
81	79
85	63
75	80
69	61
80	69
76	72
79	80
74	65

Unpaired sample test

- **Statement of hypothesis**

Null hypothesis $H_0 : \mu_1 = \mu_2$

Alternate hypothesis $H_1 : \mu_1 \neq \mu_2$

μ_1 = average percent dissolution of formulation A

μ_2 = average percent dissolution of formulation B

Null hypothesis = *there is no difference in dissolution of both formulation*

Alternate hypothesis = there is difference in dissolution of both formulation

Unpaired sample test

A X_1	B X_2	$(X_1-\bar{X}_1)^2$	$(X_2-\bar{X}_2)^2$
68	74	82.81	6.76
84	71	47.61	0.16
81	79	15.21	57.76
85	63	62.41	70.56
75	80	4.41	73.96
69	61	65.61	108.16
80	69	8.41	5.76
76	72	1.21	0.36
79	80	3.61	73.96
74	65	9.61	40.96
$\sum X_1 = 771$	$\sum X_2 = 714$	$\sum(X_1-\bar{X}_1)^2 = 300.9$	$\sum(X_2-\bar{X}_2)^2 = 438.4$
$\bar{X}_1 = 77.1$	$\bar{X}_2 = 71.4$		
$S_1^2 = 33.4333$	$S_2^2 = 48.7111$		

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_p^2}{N_1} + \frac{s_p^2}{N_2}}}$$

$$= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$$

$$S^2 = \frac{\sum(X-\bar{X})^2}{N-1}$$

$$S = \sqrt{\frac{\sum(X-\bar{X})^2}{N-1}}$$

$$S_p^2 = \frac{(N_1-1)S_1^2 + (N_2-1)S_2^2}{N_1+N_2-2}$$

$$S_p = \sqrt{\frac{(N_1-1)S_1^2 + (N_2-1)S_2^2}{N_1+N_2-2}}$$

Unpaired sample test

$$S_p = \sqrt{\frac{(N_1-1)S_1^2 + (N_2-1)S_2^2}{N_1+N_2-2}}$$

$$= \sqrt{\frac{(10-1)33.4333 + (10-1)48.7111}{10+10-2}}$$

$$= \sqrt{\frac{300.8997 + 438.3999}{18}}$$

$$= \sqrt{41.0722}$$

$$= 6.4088$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{N_1}+\frac{1}{N_2}}}$$

$$= \frac{(77.1 - 71.4) - (0)}{6.4088\sqrt{\frac{1}{10}+\frac{1}{10}}}$$

$$= \frac{5.7}{6.4088\sqrt{0.2}}$$

$$= \frac{5.7}{2.8661} = 1.988$$

$H_0 : \mu_1 = \mu_2$

Unpaired sample test

- Calculated $t = 1.988$

- The value of t obtained from **t distribution table**
  two-sided test,
  with 5% ($\alpha = 0.05$) level,
  with d.f. 18 ($N_1 + N_2 - 2$) is = 2.10

- Here, calculated t < table t value
- i.e. accept null hypothesis
- Conclusion is there is no difference in dissolution of both formulation

Unpaired sample test

The nicotine content in (milligrams) of two samples of tobacco was found to be as follows. Can it said that the two samples came from the normal distribution? **05**

($t_{9,0.05} = 2.262$ ~~$, F_{5,4,0.05} = 6.26$~~)

Sample A	24	27	26	21	25	
Sample B	27	30	28	31	22	36

Null hypothesis $H_0 : \mu_1 = \mu_2$

Alternate hypothesis $H_1 : \mu_1 \neq \mu_2$

μ_1 = average content of nicotine in sample A

μ_2 = average content of nicotine in sample B

Null hypothesis = *there is no difference , i.e. two samples have been came from the normal distribution.*

Alternate hypothesis = there is difference,

Unpaired sample test

A X_1	B X_2	$(X_1 - \bar{X}_1)^2$	$(X_2 - \bar{X}_2)^2$
24	27	0.36	4
27	30	5.76	1
26	28	1.96	1
21	31	12.96	4
25	22	0.16	49
	36		49
$\sum X_1 = 123$	$\sum X_2 = 174$	$\sum (X_1 - \bar{X}_1)^2 = 21.2$	$\sum (X_2 - \bar{X}_2)^2 = 108$
$\bar{X}_1 = 24.6$	$\bar{X}_2 = 29$		
$S_1^2 = 5.3$	$S_2^2 = 21.6$		

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_p^2}{N_1} + \frac{S_p^2}{N_2}}}$$

$$= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$$

$$S^2 = \frac{\sum (X - \bar{X})^2}{N - 1}$$

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{N - 1}}$$

$$S_p^2 = \frac{(N_1 - 1)S_1^2 + (N_2 - 1)S_2^2}{N_1 + N_2 - 2}$$

$$S_p = \sqrt{\frac{(N_1 - 1)S_1^2 + (N_2 - 1)S_2^2}{N_1 + N_2 - 2}}$$

Unpaired sample test

$$S_p = \sqrt{\frac{(N_1-1)S_1^2 + (N_2-1)S_2^2}{N_1+N_2-2}}$$

$$= \sqrt{\frac{(5-1)5.3 + (6-1)21.6}{5+6-2}}$$

$$= \sqrt{\frac{21.2 + 108}{9}}$$

$$= \sqrt{14.3556}$$

$$= 3.7889$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{N_1}+\frac{1}{N_2}}}$$

$$= \frac{(24.6 - 29) - (0)}{3.7889\ \sqrt{\frac{1}{5}+\frac{1}{6}}}$$

$$= \frac{-4.4}{3.7889\ \sqrt{0.3667}}$$

$$= \frac{-4.4}{2.2944}$$

$$= -1.9177$$

Unpaired sample test

- Calculated t = -1.9177 i.e. Value is 1.9177

- The value of t obtained from **t distribution table**
  two-sided test,
  with 5% ($\alpha = 0.05$) level,
  with d.f. 9 ($N_1 + N_2 - 2$) is = 2.2622

- negative sign of t value direction on left tailed so calculated t value concluded as 1.9177
- Here, calculated t < table t value , hence it is not significant
- i.e. accept null hypothesis
- Conclusion is *there is no difference , i.e. two samples have been came from the normal distribution with same mean*

Unpaired sample test

Synonym:

dependent sample t-test,
Repeated measures t-test,
Difference t-Test

- Test for Comparison of Means of Related Samples

- In clinical studies, a paired design is often described as one in which each patient acts as his or her own "control."
- Use the same subject as control as well as the test

- In a paired sample t-test, each subject or entity is measured twice, resulting in pairs of observations.

Paired sample test

EXCERSIZE

- A new formulation of a marketed drug is to be tested for bioavailability, comparing the extent of absorption to the marketed form on six laboratory animals. Each animal received both formulations in random order on two different occasions. The results, the area under the blood level versus time curve (AUC), are shown in Table. The significance level is set at 5%.

Results of a Bioavailability Study Comparing a New Formulation (*A*) to a Marketed Form (*B*) with Regard to the Area Under the Blood-Level Curve

Animal	*A*	*B*
1	136	166
2	168	184
3	160	193
4	94	105
5	200	198
6	174	197

Paired sample test

- statement of hypothesis

Null hypothesis $H_0 : \mu_1 = \mu_2$

Alternate hypothesis $H_1 : \mu_1 \neq \mu_2$

$$\Delta = \mu_1 - \mu_2 = 0$$

μ_1 = average of AUC of formulation A

μ_2 = average of AUC of formulation B

Null hypothesis = *there is no difference in bioavailability* of both Formulation

Alternate hypothesis = there is difference in *bioavailability* of both Formulation

$$t = \frac{\bar{\delta} - \Delta}{S/\sqrt{N}} \quad or \quad t = \frac{\bar{\delta} - (\mu_1 - \mu_2)}{S/\sqrt{N}} \quad t = \frac{\bar{d}}{\sqrt{S^2/N}} \quad t = \frac{\overline{X_d}}{S_d/\sqrt{N}}$$

Paired sample test

Animal	A	B	Difference δ	$(\delta-\bar{\delta})^2$
1	136	166	30	132.25
2	168	184	16	6.25
3	160	193	33	210.25
4	94	105	11	56.25
5	200	198	-2	420.25
6	174	197	23	20.25
			$\sum\delta = 111$	$\sum(\delta-\bar{\delta})^2 = 845.5$
			$\bar{\delta} = 18.5$	

$$S = \sqrt{\frac{\sum(X-\bar{X})^2}{N-1}} = \sqrt{\frac{\sum(\delta-\bar{\delta})^2}{N-1}} \quad \text{(In this case)}$$

$$t = \frac{\bar{\delta}-\Delta}{S/\sqrt{N}} = \frac{18.5-0}{13.0038/\sqrt{6}} = \frac{18.5}{13.0038/2.4495} = \frac{18.5}{13.0038/2.4495} = 3.4848$$

Paired sample test

- Calculated $t = 3.48$

- The value of t obtained from **t distribution table**
  two-sided test,
  with 5% ($\alpha = 0.05$) level,
  with d.f. 5 (N-1) is = 2.57

- Here, calculated t > table t value , hence difference is significant
- i.e. reject null hypothesis
  and
  accept alternative hypothesis
- Conclusion is there significant difference in *bioavailability* of both formulation

Paired sample test

EXCERSIZE

- A certain stimulant drug is administered to each of 12 patients resulted in the following change in blood pressure 5, 2, 8, -1, 3, 0, -2, 1, 5, 0, 4, 6
  Can it be concluded that the stimulant will in general be accompanied by an increase in blood pressure with ($\alpha = 0.05$)

patient	b.p. before drug	b.p. after drug	Difference δ
1			5
2			2
3			8
4			-1
5			3
6			0
7			-2
8			1
9			5
10			0
11			4
12			6

Paired sample test

- statement of hypothesis

Null hypothesis $H_0 : \mu_1 = \mu_0$

$$\Delta = \mu_1 - \mu_2 = 0$$

Alternate hypothesis $H_1 : \mu_1 > \mu_0$

μ_1 = average of blood pressure after drug administration

μ_0 = average of blood pressure before drug administration

Null hypothesis = *there is no difference in b.p. before and after drug administration*

Alternate hypothesis = *drug administration increase b.p.*

Paired sample test

Difference δ	$(\delta-\bar{\delta})^2$
5	5.8403
2	0.3403
8	29.3403
-1	12.8403
3	0.1736
0	6.6736
-2	21.0069
1	2.5069
5	5.8403
0	6.6736
4	2.0069
6	11.6736
$\sum\delta = 31$	$\sum(\delta-\bar{\delta})^2 = 104.9167$
$\bar{\delta} = 2.5833$	

$$S = \sqrt{\frac{\sum(\delta-\bar{\delta})^2}{N-1}} = \sqrt{\frac{104.9167}{12-1}} = 3.0883$$

$$\Delta = \mu_1 - \mu_2 = 0$$

$$t = \frac{\bar{\delta}-\Delta}{S/\sqrt{N}}$$

$$= \frac{2.5833-0}{3.0883/\sqrt{12}}$$

$$= \frac{2.5833}{3.0883/3.4641} = 2.8976$$

Paired sample test

- Calculated $t = 2.8976$

- The value of t obtained from **t distribution table**
  one-sided test,
  with 5% ($\alpha = 0.05$) level,
  with d.f. 11 (N-1) is =1.7959

- Here, calculated t > table t value , hence difference is significant

- i.e. reject null hypothesis
  and
  accept alternative hypothesis

- Conclusion is stimulant *drug administration increase b.p.*

ANOVA

Full form : Analysis of variance

Defination :

ANOVA is a statistical technique specially designed to test whether the means of more than 2 quantitative populations are equal.

Assumptions :

- The observations are independent
- Parent populations from which observations are taken are normal
- Various treatment and environmental effects are additive in nature

ANOVA

t test vs ANOVA

- Z test & *t* test : test the significance of the difference btwn one or two sample means.
- ANOVA: test the significance of the difference btwn more than two sample means.
- *t* test is a special case of ANOVA in which only two means are compared.

- ANOVA is a generalization of the *t* test to three or more groups.

- In t test when you reject the null hypothesis of equality of means, the alternative tells you which one of the two means is greater ($\mu_1 > \mu_2$, $\mu_1 < \mu_2$).
- With ANOVA, the corresponding test is an F test. It tells you that the means are different but not necessarily which one is larger than the others.

ONE-WAY ANOVA

Synonym: : "completely randomized" design , single factor ANOVA

The one-way ANOVA is an extension of the two independent groups t test when an experiment contains more than two treatments

- One way Analysis of Variance (ANOVA) is a hypothesis test in which only one categorical variable or single factor is considered.

- Null hypothesis (H_0) is the equality in all population means,

$$H_0: \mu_1 = \mu_2 = \mu_3 \ldots = \mu_k = \mu$$

- Alternative hypothesis (H_1) is the difference in at least one mean.

Steps of ANOVA calculation

EXCERSIZE

15 tablets are tested for the comparison of three assay methods, 5 tablets for each assay.

Results of Assays Comparing Three Analytical Methods

Method *A*	Method *B*	Method *C*
102	99	103
101	100	100
101	99	99
100	101	104
102	98	102

the objective of this experiment is to compare the average results of the three methods.

ONE-WAY ANOVA

- statement of hypothesis

Null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3$

Alternate hypothesis $H_1 : \mu_x \neq \mu_y$

$\mu_1 =$ average assay of tablets using method A
$\mu_2 =$ average assay of tablets using method B
$\mu_3 =$ average assay of tablets using method C
μ_x & μ_y for any two means

Null hypothesis = *there is no difference in all three assay methods*

Alternate hypothesis = at least two assay methods are different , not all the assay methods are equal

ONE-WAY ANOVA

method A Y_A	method B Y_B	method C Y_C	Y_A^2	Y_B^2	Y_C^2
102	99	103	10404	9801	10609
101	100	100	10201	10000	10000
101	99	99	10201	9801	9801
100	101	104	10000	10201	10816
102	98	102	10404	9604	10404
$\sum Y_A = 506$	$\sum Y_B = 497$	$\sum Y_C = 508$	$\sum Y_A^2 = 51210$	$\sum Y_B^2 = 49407$	$\sum Y_C^2 = 51630$
$\sum Y = G = \sum Y_A + \sum Y_B + \sum Y_C = 1511$			$\sum Y^2 = \sum Y_A^2 + \sum Y_B^2 + \sum Y_C^2 = 1,52,247$		

$$CT = \frac{(\sum Y)^2}{N} = \frac{G^2}{N}$$

Where, $T_A = \sum Y_A$, $T_B = \sum Y_B$, $T_C = \sum Y_C$

$$TSS = \sum Y^2 - CT \qquad BSS = \sum \frac{T_i^2}{N_i} - CT$$

Calculation steps

Step 1. compute G

- Grand total of all observations

G= 1511

Step 2. compute correction term (CT) / correction factor (CF)

$$CT = \frac{(\sum Y)^2}{N} = \frac{G^2}{N} = \frac{1511^2}{15} = 152208.0667$$

Step 3. compute raw sum of square

- sum of the square of all observations

$$\sum Y^2 = \sum Y_A^2 + \sum Y_B^2 + \sum Y_C^2 = 1{,}52{,}247$$

Calculation steps

Step 4. compute total sum of square (TSS)

$$TSS = \sum Y^2 - CT$$

$$= 152247 - 152208.0667$$

$$= 38.9333$$

Step 5. compute between sum of square (BSS)

$$BSS = \sum \frac{T_i^2}{N_i} - CT = \frac{T_A^2}{N_A} + \frac{T_B^2}{N_B} + \frac{T_C^2}{N_C} - CT$$

Where, $T_A = \sum Y_A$, $T_B = \sum Y_B$, $T_C = \sum Y_C$

$$= \frac{506^2}{5} + \frac{497^2}{5} + \frac{508^2}{5} - 152208.0667$$

$$= 152221.8 - 152208.0667$$

$$= 13.7333$$

Calculation steps

Step 6. compute within sum of square (WSS)

WSS = TSS - BSS
= 38.9333 – 13.7333
= 25.2

Step 7. compute degree of freedom

Total d.f. = N-1 = 15-1 = 14
Between- method d.f = k -1 = 3 -1 = 2
Within- method t d.f. = N-k = 15-3 =12

Where, N = total observations
k = total groups

Calculation steps

Step 8. compute between classes Mean sum of square (MBSS)

$$\text{MBSS} = \frac{BSS}{Between-method\ d.f} = \frac{13.7333}{2} = 6.8667$$

Step 9. compute within classes Mean sum of square (MWSS)

$$\text{MWSS} = \frac{WSS}{Within-method\ d.f.} = \frac{25.2}{12} = 2.1$$

Step 10. compute calculated F value, Variance ratio

$$\text{F} = \frac{MBSS}{MWSS} = \frac{6.8667}{2.1} = 3.2698$$

Calculation steps

ANOVA table

Source of variation	d.f.	Sum of square	Mean square	F
BSS	2	13.7333	6.8667	3.2698
WSS	12	25.2000	2.1000	
TSS	14	38.9333		

Step 11. find critical value of F using a table of the *F* distribution

with 5% ($\alpha = 0.05$) level,

with 2 d.f. in the *numerator*

and

12 d.f. in the *denominator* = *3.885*

ONE-WAY ANOVA

- Here, calculated F < table F value , hence no significant difference
- i.e. accept null hypothesis

Step 12. Conclusion

all means are equal: method A = method B = method C

There is no difference in all three methods

ONE-WAY ANOVA

EXCERSIZE

- It is suspected that four analysts in a laboratory are not performing accurately. A known sample is given to each analyst and replicate assays performed by each with the following results:
- Perform an ANOVA. significant level is 5% level

Analyst			
I	II	III	IV
10	9	8	9
11	10	9	9
10	11	8	8

ONE-WAY ANOVA

- statement of hypothesis

Null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$

Alternate hypothesis $H_1 : \mu_x \neq \mu_y$

$\mu_1 =$ average assay of samples by analyst I
$\mu_2 =$ average assay of samples by analyst II
$\mu_3 =$ average assay of samples by analyst III
$\mu_4 =$ average assay of samples by analyst IV

μ_x & μ_y for any two means

Null hypothesis = *there is no difference in all four analyst assay*

Alternate hypothesis = at least two analyst assay are different

ONE-WAY ANOVA

I Y_I	II Y_{II}	III Y_{III}	IV Y_{IV}	Y_I^2	Y_{II}^2	Y_{III}^2	Y_{IV}^2
10	9	8	9	100	81	64	81
11	10	9	9	121	100	81	81
10	11	8	8	100	121	64	64
$T_I = 31$	$T_{II} = 30$	$T_{III} = 25$	$T_{IV} = 26$				
$G = 112$				$\sum Y^2 = 1058$			

$$CT = \frac{G^2}{N} = \frac{112^2}{12} = 1045.3333$$

ONE-WAY ANOVA

$$TSS = \sum Y^2 - CT$$

$$= 1058 - 1045.3333$$

$$= 12.6667$$

$$BSS = \sum \frac{T_i^2}{N_i} - CT$$

$$= \frac{31^2}{3} + \frac{30^2}{3} + \frac{25^2}{3} + \frac{26^2}{3} - 1045.3333$$

$$= 1054 - 1045.3333$$

$$= 8.6667$$

$$WSS = TSS - BSS$$

$$= 12.6667 - 8.6667$$

$$= 4$$

ONE-WAY ANOVA

$$\text{Total d.f.} = \text{N-1} = 12\text{-}1 = 11$$

$$\textit{Between- analyst d.f} = \text{k -1} = 4\text{ -1} = 3$$

$$\textit{Within- analyst d.f.} = \text{N-k} = 12\text{-}4 = 8$$

$$\text{MBSS} = \frac{BSS}{Between-analyst\ d.f} = \frac{8.6667}{3} = 2.8889$$

$$\text{MWSS} = \frac{WSS}{Within-analyst\ d.f.} = \frac{4}{8} = 0.5$$

$$\text{F} = \frac{MBSS}{MWSS} = \frac{2.8889}{0.5} = 5.7778$$

ONE-WAY ANOVA

ANOVA table

Source of variation	d.f.	Sum of square	Mean square	F
BSS	3	8.6667	2.8889	5.7778
WSS	8	4	0.5	
TSS	11	12.6667		

finding critical value of F using a table of the *F* distribution

with 5% ($\alpha = 0.05$) level,

with 3 d.f. in the *numerator*

and

8 d.f. in the *denominator* $= 4.066$

- Here, calculated F > table F value , hence significant difference
- i.e. Reject null hypothesis , thus accept alternative hypothesis

Conclusion - at least two analyst assay are different

LSD test

Full form : Least Significant Difference

when ANOVA results in rejection of the null hypothesis,

- ANOVA gives significant result as at least one of the groups tested differs from the other groups. However, you can't tell from the ANOVA test which group differs.

- In order to find out exactly which groups are different from each other, we must conduct a post-hoc test.
- One commonly used post-hoc test is *Fisher's least significant difference test.*

- This test is a simple *t test* comparing all possible pairs of treatment means.

LSD test

- The LSD calculates the smallest significant between two means as if a test had been run on those two means.
- Any difference larger than the LSD is considered a significant result.

$$\text{LSD} = t\sqrt{S^2\left(\frac{1}{N_1}+\frac{1}{N_2}\right)}$$

Where,

t = the tabulated value of t at the α level, with d.f. from the WMSS of the ANOVA table

S^2 = the within mean square variance= WMSS

N_1 and N_2 sample size

If the sample sizes are equal in each group ($N_1 = N_2 = N$), $\text{LSD} = t\sqrt{\frac{2s^2}{N}}$

LSD test

- Finding LSD in EXCERSIZE 2

- t = 2.306
- $S^2 = 4$

Analyst			
I	II	III	IV
10	9	8	9
11	10	9	9
10	11	8	8

ANOVA table

Source of variation	d.f.	Sum of square	Mean square	F
BSS	3	8.6667	2.8889	5.7778
WSS	8	4	0.5	
TSS	11	12.6667		

$$LSD = t\sqrt{S^2\left(\frac{1}{N_1}+\frac{1}{N_2}\right)}$$

$$= 2.306\sqrt{4\left(\frac{1}{3}+\frac{1}{3}\right)} \quad = 1.33137$$

LSD test

LSD = 1.33137

	I Y_I	II Y_{II}	III Y_{III}	IV Y_{IV}
	10	9	8	9
	11	10	9	9
	10	11	8	8
SUM	31	30	25	26
MEAN	10.3333	10.0000	8.3333	8.6667
		$\bar{Y}_I-\bar{Y}_{II}$= 0.33	$\bar{Y}_I-\bar{Y}_{III}$= 2	$\bar{Y}_I-\bar{Y}_{IV}$= 1.67
			$\bar{Y}_{II}-\bar{Y}_{III}$= 1.67	$\bar{Y}_{II}-\bar{Y}_{IV}$= 1.33
				$\bar{Y}_{III}-\bar{Y}_{IV}$= -0.33

The mean difference , grater than LSD in red colour

Conclusion - *I* differs from *III*, and *IV*;
II differs from *III*, and *IV*

TWO-WAY ANOVA

Synonym: : "randomized block design, two factor ANOVA

The two-way ANOVA is an extension of the paired t test to more than two treatments.

- Two way Analysis of Variance (ANOVA) is a hypothesis test in which two categorical variable or two factors are considered.

- A two-way ANOVA tests the effect of two independent variables on a dependent variable

EXCERSIZE

A Comparison of Dissolution of two generic products and a standard drug Tablets Formulations in Eight laboratories ($F_{2,14}=3.74$, $F_{7,14}=2.14$)

Tablet Dissolution After 30 Minutes for Three Products (Percent Dissolution)

	Generic		
Laboratory	***A***	***B***	**Standard**
1	89	83	94
2	93	75	78
3	87	75	89
4	80	76	85
5	80	77	84
6	87	73	84
7	82	80	75
8	68	77	75

TWO-WAY ANOVA

- statement of hypothesis two categories of both hypothesis

for differences among tablet formulations

Null hypothesis $H_{0C} : \mu_A = \mu_B = \mu_S$

Alternate hypothesis $H_{1C} : \mu_x \neq \mu_y$

$\mu_A =$ average of dissolution by product A
$\mu_B =$ average of dissolution by product B
$\mu_S =$ average of dissolution by standard

μ_x & μ_y for any two means

Null hypothesis = *there is no difference in all three tablet formulations*

Alternate hypothesis = at least two *formulations* are different

TWO-WAY ANOVA

for differences among laboratory

Null hypothesis $H_{0R} : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6 = \mu_7 = \mu_8$

Alternate hypothesis $H_{1R} : \mu_x \neq \mu_y$

μ_1 to μ_8 = average of dissolution of products from particular laboratories

μ_x & μ_y for any two means

Null hypothesis = *there is no difference in all laboratories*

Alternate hypothesis = at least two *laboratories* are different

TWO-WAY ANOVA

A Y_A	B Y_B	Standard Y_S	Row Sum	Y_A^2	Y_B^2	Y_S^2
89	83	94	266	7921	6889	8836
93	75	78	246	8649	5625	6084
87	75	89	251	7569	5625	7921
80	76	85	241	6400	5776	7225
80	77	84	241	6400	5929	7056
87	73	84	244	7569	5329	7056
82	80	75	237	6724	6400	5625
68	77	75	220	4624	5929	5625
C_A = $\sum Y_A$= 666	C_B = $\sum Y_B$= 666616	C_S = $\sum Y_S$= 664				
G = 1946				$\sum Y^2$ = 158786		

$$CT = \frac{G^2}{N} = \frac{1946^2}{24} = 157788.1667$$

$$TSS = \sum Y^2 - CT$$
$$= 158786 - 157788.1667$$
$$= 997.8333$$

Column sum of squares (CSS) or product SS

$$CSS = \sum \frac{C_i^2}{R} - CT$$

$$= \frac{666^2}{8} + \frac{616^2}{8} + \frac{664^2}{8} - 157788.1667$$

$$= 200.333$$

$$\text{Row sum of squares (RSS)} = \sum \frac{R_i^2}{C} - CT$$

$$= \frac{266^2}{3} + \frac{246^2}{3} + \frac{251^2}{3} + \frac{241^2}{3} + \frac{241^2}{3} + \frac{244^2}{3} + \frac{237^2}{3} + \frac{220^2}{3} - 157788.1667$$

$$= 391.8333333$$

Residual ($C \times R$) sum of squares (ESS) = TSS - CSS – RSS

(Sum of squares due to error) = 997.8333 -200.333-391.833

= 405.6667

TWO-WAY ANOVA

Total d.f. = N-1 = 24-1 = 23

Column d.f = C -1 = 3 -1 = 2

Row d.f. = R-1 = 8 -1 = 7

Residual *d.f.* = (C -1) * (R -1) = 2 * 7 = 14

$$\text{MCSS} = \frac{CSS}{Column\ d.f} = \frac{200.333}{2} = 100.1667$$

$$\text{MRSS} = \frac{RSS}{Row\ d.f.} = \frac{391.833}{7} = 55.9762$$

$$\text{MESS} = \frac{ESS}{Residual\ d.f.} = \frac{405.6667}{14} = 28.9762$$

TWO-WAY ANOVA

$$F_C = \frac{MCSS}{MESS} = \frac{2100.1667}{28.9762} = 3.4569$$

$$F_R = \frac{MRSS}{MESS} = \frac{55.9762}{28.9762} = 1.9318$$

ANOVA table

Source of variation	d.f.	Sum of square	Mean square	F
Columns	2	200.3333	100.1667	F_C =3.46 F_R =1.93
Row	7	391.8333	55.9762	
Residual (C ×R)	14	405.6667	28.9762	
Total	23	998		

TWO-WAY ANOVA

- To test for differences among tablet formulations

finding critical value of F_C using a table of the *F* distribution
with 5% ($\alpha = 0.05$) level,
with 2 d.f. in the *numerator*
and
14 d.f. in the *denominator* = *3.739*

- Here, calculated F < table F value , hence no significant difference
- i.e. accept null hypothesis H_{0C}

Conclusion - *there is no difference in all three tablet formulations*

TWO-WAY ANOVA

- To test for differences among laboratory

finding critical value of F_R using a table of the *F* distribution
with 5% ($\alpha = 0.05$) level,
with 7 d.f. in the *numerator*
and
14 d.f. in the *denominator* = *2.764*

- Here, calculated F < table F value , hence no significant difference
- i.e. accept null hypothesis H_{0R}

Conclusion - *there is no difference in all laboratories*

comparisons of each generic product versus the standard

- Use LSD

$$\text{LSD} = t\sqrt{S^2\left(\frac{1}{N_1}+\frac{1}{N_2}\right)}$$

t = the tabulated value of t at the α level, with d.f. from the MESS of the ANOVA table

S^2 = the within mean square variance= MESS

N_1 and N_2 = size of observation in particular groups

- t = 2.1448
- S^2 = 28.9762

TWO-WAY ANOVA

$$LSD = t\sqrt{S^2\left(\frac{1}{N_1}+\frac{1}{N_2}\right)} = 2.1448\sqrt{28.9762\left(\frac{1}{8}+\frac{1}{8}\right)}$$

$$= 5.7727$$

	A Y_A	B Y_B	Standard Y_S
MEAN	83.25	77	83
		$\bar{Y}_A-\bar{Y}_B=$ 6.25	$\bar{Y}_A-\bar{Y}_S=$ 0.25
			$\bar{Y}_B-\bar{Y}_S=$ -6

Conclusion – *formulation B differs* **from** *standard*
formulation A differs **from** *B*

ONE-WAY *vs* TWO-WAY

	One-way ANOVA	Two-way ANOVA
Deals with…	only one categorical variable	two categorical variables
Synonym	completely randomize design	randomized block design
Extension of…	extension of the unpaired t test	extension of the paired t test
Hypothesis	One null hypothesis One alternative hypothesis	Two null hypothesis: 1. The means of observation by one variable is the same. 2. The means of observation by the other variable is the same. Two alternative hypothesis
Compute…	total sum of square (TSS) between sum of square (BSS) within sum of square (WSS)	total sum of square (TSS) Row sum of square (RSS) column sum of square (CSS) Residual sum of square (ESS)
F value	One value for one variable	Two F values for two variables

Study questions

Two types of drug were used on 5 and 7 patients for reducing their weight. Drug A is imported and drug B indigenous. The decrease in weight after using the drugs for six months was recorded as given below. Is there significant difference in efficiency of two drugs? If not which drug should be used ? ($t_{10,0.05} = 2.225$) **05**

Imported A	11	13	12	14	10		
Indigenous B	12	9	8	15	14	9	10

WINTER • 2014

Ans. $H_0 : \mu_A = \mu_B$, $H_1 : \mu_A \neq \mu_B$, $t = 0.735$, not significant

Since drug B is indigenous and there is no difference in efficiency between the two with no side effects, we should buy the Herbal drug.

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$$

Study questions

Group of 5 and 7 patients treated with medicine , for weight of patient data given below, $\alpha = 0.05$

Weight of patient treated with Medicine A	Weight of patient treated with Medicine B
42	38
39	42
48	56
60	64
41	68
	69
	62

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$$

Do you agree with the claim that medicine B increases the weight significantly ?

Ans. $H_0 : \mu_A = \mu_B$, $H_1 : \mu_A < \mu_B$, $t = -1.7$, accept H_0

Study questions

The following data show the blood pressure reduction (in mm Hg) caused in 10 animals by a new antihypertensive compound. **05**

20	18	15	12	8	16	18	17	14	21

Test the Hypothesis that the blood pressure reduction for the population is 15 mm Hg. ($t_{9,0.05} = 2.262$)

Ans. $H_0 : \mu_1 = \mu_0$, $H_1 : \mu_1 \neq \mu_0$, $t = 0.74$, accept H_0

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$

Measurement of body mass index(BMI) for a sample of 10 healthy adult males are shown in the following table.

Subject	1	2	3	4	5	6	7	8	9	10
BMI	21	23	32	24	47	22	45	37	24	35

On the basis of these data can we conclude that the mean BMI of the population from the sample was drawn is 35? $T_{tab}(9,0.05) = 2.262$

Ans. $H_0 : \mu_1 = \mu_0$, $H_1 : \mu_1 \neq \mu_0$, $t = -1.3031$, accept H_0

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$

Study questions

A random sample of 20 famotidine injection from a batch gives a mean active ingredient content of 42 mg and the standard deviation of 5 mg. Test the hypothesis that the population mean is 40 mg. ($t_{19,0.05} = 2.09$) **05**

$$t = \frac{|\overline{X} - \mu_0|}{S/\sqrt{N}}$$

Thirty microgram of Methcobalamin was given intramuscularly every four week to six patients of pernicious anaemia. The results are given below. Do the data indicate improvement in haemoglobin level? ($t_{5,\,0.05} = 2.57$) **06**

Patient	Haemoglobin gm %	
	Before therapy	After 3 months therapy
1	10.3	13.0
2	11.3	13.4
3	14.7	16.0
4	11.3	13.6
5	11.7	14.0
6	12.5	13.8

$$t = \frac{\overline{\delta} - \Delta}{S/\sqrt{N}}$$

Study questions

The following are the results of assays comparing three analytical methods:

Method A	Method B	Method C
100	100	101
102	99	100
99	101	101
104	98	102
101	98	100

Test at 5% level of significance the null hypothesis that there are no significant differences among the three methods. ($F_{T\,(2,\,12\,=\,3.88)}$)

Ans.

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*
Between Groups	11.2	2	5.6	2.754098
Within Groups	24.4	12	2.033333	
Total	35.6	14		

Study questions

In preclinical study animal were treated with two antihypertensive experimental drugs and control with 12 animal randomly assigned to three groups, four per group. One animal died from nondrug related cause and was lost during experimentation. Changes in pressure from base line are shown in table. Perform the one way ANOVA analysis and find out the treatment difference and inherent variation in animal group response ($F_{2,8}$=4.46).

Drug - 1	15	12	19	11
Drug - 2	8	14	13	6
Control	-	16	20	22

Ans.

ANOVA				
Source of Variation	*SS*	*df*	*MS*	*F*
Between Groups	141.4697	2	70.73485	5.538781
Within Groups	102.1667	8	12.77083	
Total	243.6364	10		

Study questions

The following data gives the yields of 12 plots under 4 varieties of herbal crop. Test using one way ANOVA whether there are significant differences among the four varieties. ($F_{tab(3,8,0.05)} = 4.07$)

A	B	C	D
200	230	250	300
190	270	300	270
240	150	145	180

Ans.

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*
Between Groups	2839.58	3	946.528	0.25947
Within Groups	29183.3	8	3647.92	
Total	32022.9	11		

Study questions

The determination of C_{max} of drugs in µg/ml at three different formulations A, B and C was the subject of recent experiment. Four different subjects chosen at random from a group were used for this purpose. The data recorded are given in the following table.

Subject	A	B	C
1	29	11	15
2	17	4	9
3	34	6	27
4	40	9	25

Carry out the two – way ANOVA and test at 5% level of significance that [1] there is no significant difference among the subjects and [2] there is no significant difference among the maximum plasma concentrations of the three formulations A, B and C. [$F_{3,6,0.05}$= 4.76 and $F_{2,6,0.05}$= 5.14]

Ans.

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*
Rows	373.6667	3	124.5556	4.75
Columns	1012.667	2	506.3333	19.30932
Error	157.3333	6	26.22222	
Total	1543.667	11		

End of the topic

Reference:

Ph'ceutical statistics- Practical and clinical applications –Sanford Bolton

Fundamental of Statistics -S.C. Guptha

Net Source